1D-DCT

$$\underline{s} = \sum_{i=1}^{M} s_i \phi_i = \sum_{i=1}^{M} x_i \underline{d}_i = \sum_{i=1}^{M} \langle \underline{s}, \underline{d}_i \rangle \underline{d}_i$$

where $d_i = c_i \cos\left(i\pi \frac{2n+1}{2M} \right)$ $\qquad i, n = 1, \dots, M$

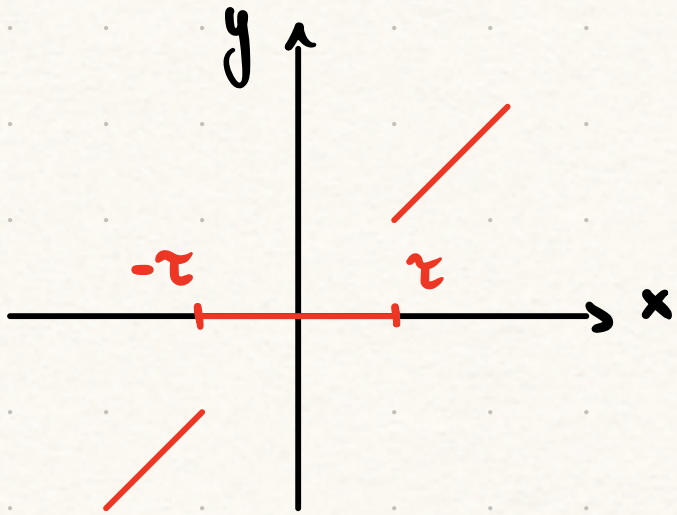
$c_1 = \sqrt{1/M}$, $c_i = \sqrt{2/M}$

$$D = \{ \underline{d}_i \}_{i=1}^{M} \qquad \underline{x} = D \underline{s}$$

$$\begin{tikzcd}
\underline{s} \arrow[r, "D^T"] & \underline{x} \arrow[d, "\text{Thresholding}"] \\
\hat{\underline{s}} & \hat{x} \arrow[l, "D"']
\end{tikzcd}$$

$$HT_\tau(x) = \begin{cases} 0 & |x| < \tau \\ x & |x| > \tau \end{cases}$$

$\rightarrow$



$s \in \mathbb{R}^M$ $\qquad \underline{s} = \underline{y} + \underline{\eta}$ $\qquad \underline{y} \in \mathbb{R}^M$ noise-free signal

$\eta \in \mathbb{R}^M$, $\eta \sim N(0, \sigma^2)$

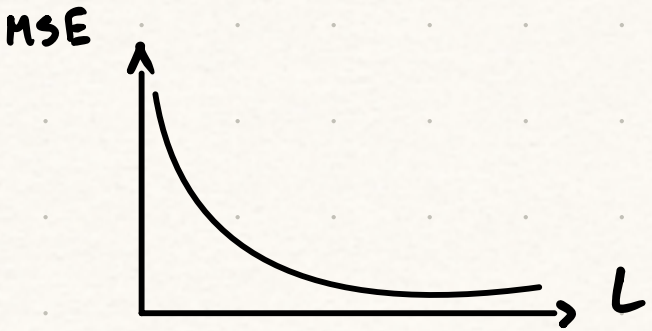
<u>Prior 1</u> $\underline{y}$ has a sparse representation in DCT

<u>Prior 2</u> $\underline{\eta}$ has not a sparse representation in DCT

To define the quality of our reconstruction we have 2 main properties

1) compression ratio $\frac{\#\text{NBC of } \hat{x}}{M}$
2) mse

If we plot MSE we would expect



All this was for the 1D signal, now we pass to Images

Images are 2D signals

$I \in [0,1]^{H \times W}$ $\qquad$ $\mathbb{R}^{H \times W}$ in general

$H = W = \sqrt{n}$

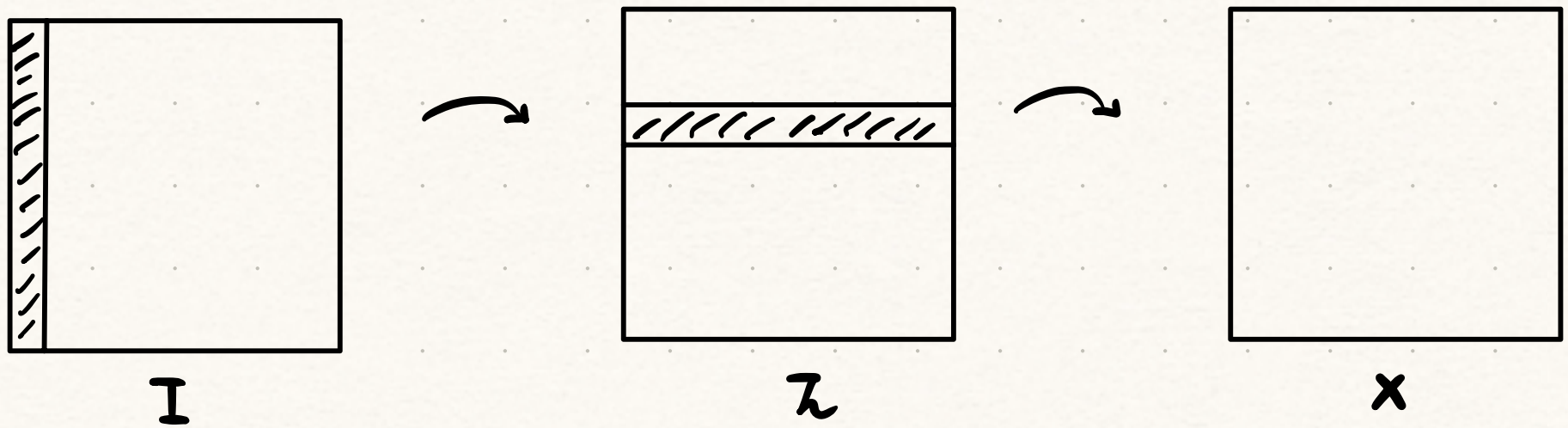
Image can be represented as $I = \sum_{m,n=1}^{\sqrt{n}} \alpha_{m,n} E_{m,n}$

$$I = \begin{bmatrix} r_1 \\ \hline r_2 \\ \hline \vdots \end{bmatrix} \Rightarrow I_v = [\, r_1 \mid r_2 \mid \cdots \,]$$

$$I \in \mathbb{R}^{\sqrt{n} \times \sqrt{n}} \longrightarrow I_v \in \mathbb{R}^n$$

This transformation does not work because we lose space order

We do:



$$Z_{i,k} = D^T I_{i,k}$$

$$Z_{i,k} = \sum_{j=1}^{\sqrt{M}} d_{j,k} I_{j,k}$$

$$X^T = D^T Z^T$$

$$\downarrow$$

$$\boxed{X = D^T I D}$$

(*)

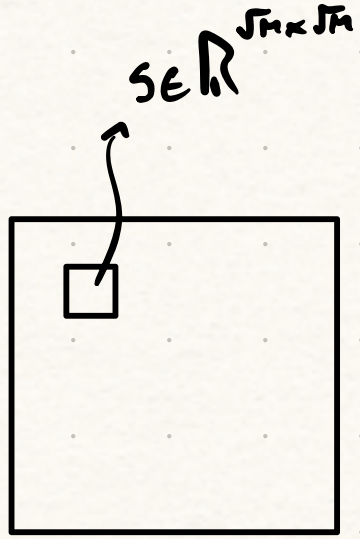
(*) is a linear transformation so it exist D_2 st:

$$\varphi(I) = D^T I D \qquad x_v = \varphi(I_v) = D_2^T I_v$$

D_2 is the Kroenecker product between two D $\qquad D_2 = D \otimes D$

In practice $I \in \mathbb{R}^{H \times W}$ where $HW \sim 10^6$

so we split the image in smaller patches



$$\begin{tikzcd}
s \arrow[r, "\text{2D-DCT}"] \arrow[dr, "\text{encoding}"'] & X \arrow[d, "HT_\tau \ \text{(tresholding)}"] \\
& \hat{X} \arrow[d, "\text{2D-DCT}"] \\
& \hat{s}
\end{tikzcd}$$

Universal treshold: $\tau = \sigma\sqrt{2\log N}$ comes from 3σ theory

(99.73 %)

$$\rightarrow \quad CR = \frac{\#NZC}{HW}$$

$$\rightarrow \quad PSNR = 10 \log_{10} \left[\frac{1}{MSE(\hat{I}, I)} \right]$$

Exercise Session 2

Sliding DCT, Wiener Filter

DCT can be used both for compression and denoising

Denoising

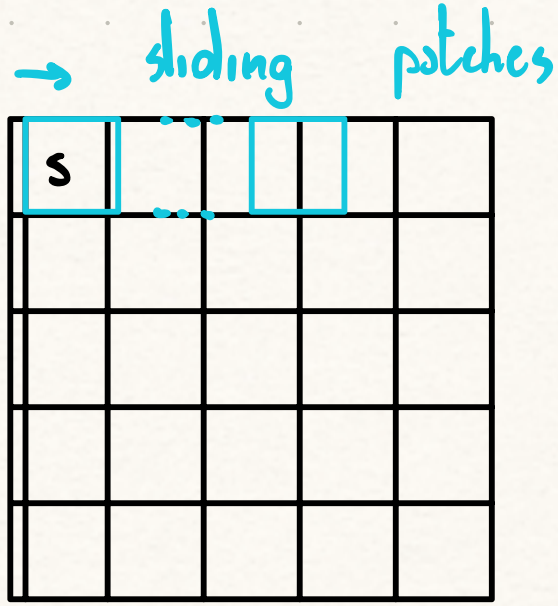
$S \in \mathbb{R}^{H \times W}$

$$s = Y + \eta$$

Y → sparse wrt D

η → dense wrt D

We want to recover Y



$$S \in \mathbb{R}^{H \times W}$$
$$Y \in \mathbb{R}^{H \times W}$$
$$\eta \sim N(0, \sigma^2 I)$$

$$S = Y + \eta$$

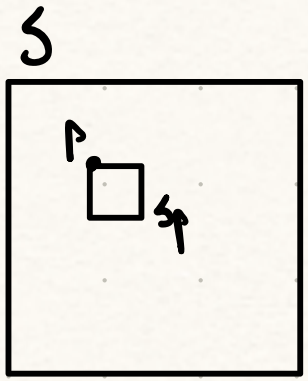
T_p is the set of patches containing p

$$\hat{Y}_{[p]} = \frac{\sum_{j \in T_p} [w_j \hat{y}_j(p)]}{\sum_{j \in T_p} w_j}$$

How to define w_j?

We can define $w_j = 1 \quad \forall j$ but we'd have more information

We can define $w_j = \dfrac{1}{\|\hat{x}_j\|_0}$ SPARSITY AWARE FILTER



$$\begin{tikzcd} s_p \arrow[r, "D^T"] & x_p \arrow[d, "\Gamma_\tau"] \\ \hat{y}_p & \hat{x}_p \arrow[l] \end{tikzcd}$$

$$\hat{Y}_w = Y_w + w_p \hat{Y}_p$$

$$Y = \frac{\hat{Y}_w}{w}$$

$$x = D^T s = \underbrace{D^T y}_{v} + D^T \eta$$

we want to approximate v as $\hat{x} = \varphi(x)$

where φ is a linear function $\varphi(x) = \alpha x$

$$MSE[\varphi(x), v] = E[(\varphi(x) - v)^2] = \dots = \alpha^2 E[x^2] + v^2 - 2\alpha v E[x]$$

$$E[x] = E[D^T y] = v \qquad E[x^2] = Var(x) + E[x]^2 = \sigma^2 + v^2$$

$$MSE(\varphi(x), v) = \alpha^2 (\sigma^2 + v^2) + v^2 - 2\alpha v^2$$

$$\alpha^* = \frac{v^2}{v^2 + \sigma^2} \qquad \hat{x}^{WE} = \frac{v^2}{v^2 + \sigma^2} x$$

Wiener Filter for denoising

Instead of v we use its HT version

$$\hat{X}^{WE}_{HT} = \frac{v_{HT}^2}{v_{HT}^2 + \sigma^2}$$

HT = Hard Tresholded

Start $\quad S = Y + \eta$

Apply sliding DCT $\quad \hat{Y}_{HT}$

From $\hat{Y}_{HT}$ take a patch $\hat{y}_{HT} \rightarrow v^{HT} = D^T \hat{y}_{HT}$

$$\hat{x}^{WE} = \frac{(v^{HT})^2}{(v^{HT})^2 + \sigma^2} x_i$$